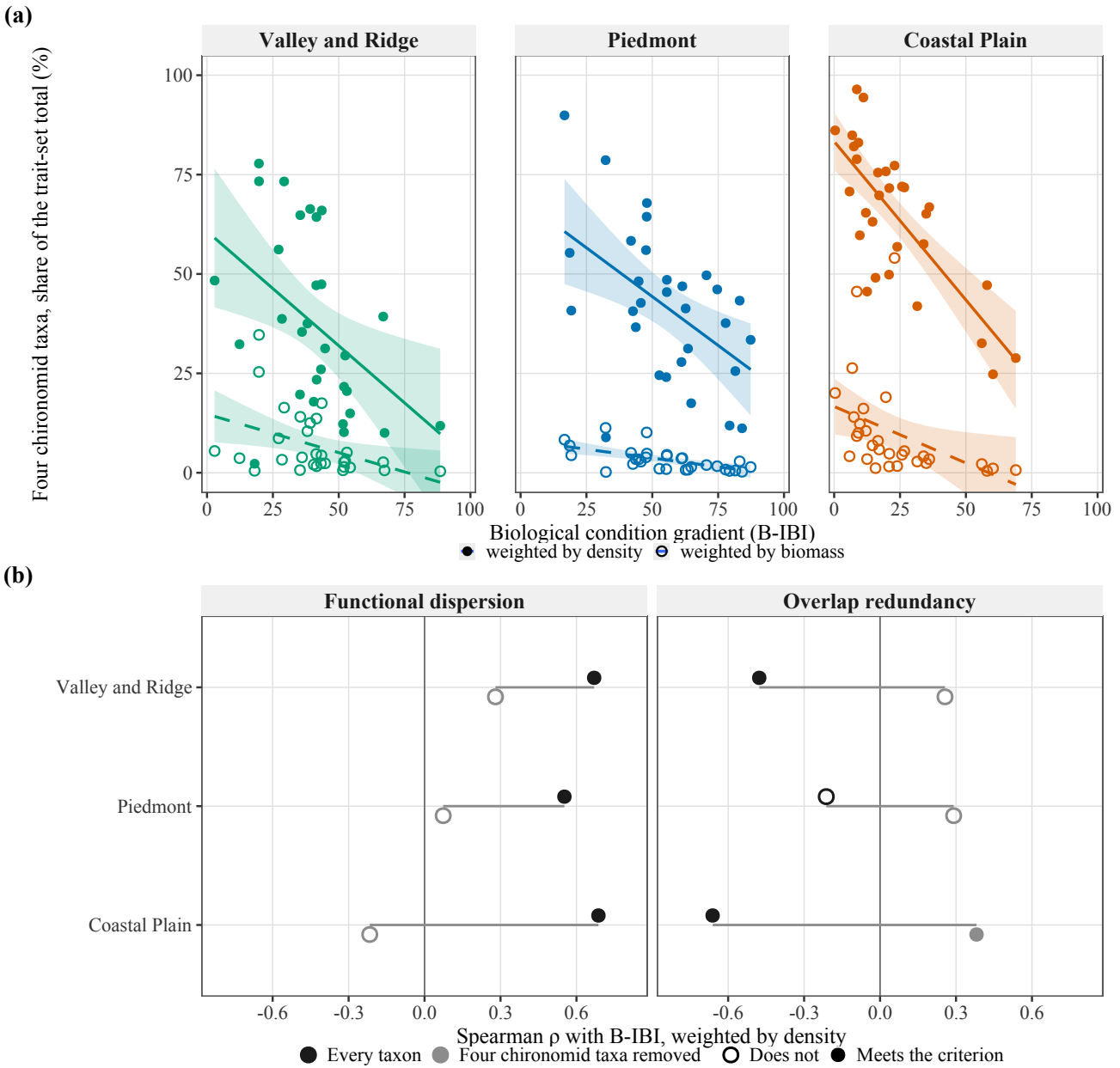

(a)
Valley and Ridge
Piedmont
Coastal Plain
Four chironomid taxa, share of the trait-set total (%)
100
75
50
25
0
0
25
50
75
100
Biological condition gradient (B-IBI)
weighted by density
weighted by biomass
(b)
Functional dispersion
Overlap redundancy
Valley and Ridge
Piedmont
Coastal Plain
-0.6
-0.3
0.0
0.3
0.6
Spearman ρ with B-IBI, weighted by density
Every taxon
Four chironomid taxa removed
Does not
Meets the criterion